

Artificial Intelligence Assignment Report 3

Warren Roberts

Student ID: 100450116

Student Email: wlr006@uark.edu

University of Arkansas

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Abstract

This report documents my implementation for Assignment 3 in the UC Berkeley Pacman framework using Python. The project extended the search algorithms from Assignment 2 to solve three more advanced tasks: visiting all four corners, designing a heuristic for the corners problem, and collecting all food in the maze. In addition, I implemented a greedy closest-food strategy for suboptimal search. A major requirement of this assignment was that Pacman must hit a wall at least once, but not more than two times. My implementation prepends a sequence of moves to perform wall-hit action before searching begins, while keeping the main search logic focused on legal state expansion.

Introduction

Assignment 3 builds directly on Assignment 2. Instead of searching for a path to one goal location, Pacman must now solve more complex objectives:

- visit all four corners in the maze,
- collect all remaining food,
- and produce a suboptimal strategy when exact search becomes expensive.

The project uses the abstract `SearchProblem` interface defined in `search.py`, including:

- `get_start_state()`
- `is_goal_state(state)`
- `get_successors(state)`
- `get_cost_of_actions(actions)`
- `is_wall(state)`

My solutions reuse the core search algorithms from Assignment 2: Depth-First Search (DFS), Breadth-First Search (BFS), Uniform Cost Search (UCS), and A^* Search. The main new work in this assignment was defining better state spaces and heuristics.

Wall-Hit Requirement

Pacman must hit a wall at least once and no more than two times to win. Since legal successor generation normally excludes walls, I handled this outside the main search tree and within the `search.py` search methods.

For `FoodSearchProblem` and `ClosestDotSearch` agent, I used special-case handling because the food-search layouts required a slightly different setup for the wall-hit behavior.

Corners Problem

The first new task in Assignment 3 was implementing `CornersProblem` in `search_agents.py`. In this problem, Pacman must collect food from all four corners of the maze.

```
macbookair:code warrenroberts$ python3 pacman.py -l medium_corners -p SearchAgent -a fn=bfs,prob=CornersProblem
[SearchAgent] using function bfs
[SearchAgent] using problem type CornersProblem
Path found with total cost of 109 in 0.0 seconds
Search nodes expanded: 1969
Your Pacman Hits Walls 1 Times
Pacman emerges victorious! Score: 432
Average Score: 432.0
Scores: 432.0
Win Rate: 1/1 (1.00)
Record: Win
macbookair:code warrenroberts$
```

Figure 1: Corners Problem Results

State Representation States could not be just a position anymore because we needed to remember what has been explored. Each visited corner was stored in the `frozenset` `visited_corners` so the search could distinguish identical positions reached with different corner histories.

current position visited corners)

where:

- current position is Pacman's (x, y) coordinate,
- visited corners is a `frozenset` containing the corners already reached.

Start State The start state returns Pacman's initial position together with the visited-corner set. If Pacman begins on a corner, that corner is immediately inserted into the visited set so the state remains correct for any valid layout.

Goal State state is a goal state when all four corners have been visited:

visited corners = 4

Successor Function The successor function considers the four compass directions: North, South, East, and West. If the resulting square is not a wall, the new position is generated. If that position is one of the four corners, it is added to the visited-corner set. Because each search state stores both position and visited-corner history, wall checks first extract the position component before testing against the maze grid.

Corners Heuristic

To improve * performance on the corners problem, I implemented `corners_heuristic(state, problem)`.

Heuristic Idea The heuristic estimates the remaining work in two parts:

1. the Manhattan distance from the current position to the nearest unvisited corner,
2. plus the cost of a minimum spanning tree (MST) over the remaining unvisited corners.

Formally:

$$h(n) = \min_c \text{Manhattan}(pos, c) + \text{MST}(R)$$

where R is the set of remaining corners.

```
macbookair:code warrenroberts python3 pacman.py -l medium_corners -p SearchAgent -a fn=bfs,prob=CornersProblem,heuristic=corne
rs_heuristic
[SearchAgent] using function bfs
[SearchAgent] using problem type CornersProblem
Path found with total cost of 189 in 0.8 seconds
Search nodes expanded: 1966
Your Pacman Hits Walls 1 Times
Pacman emerges victorious! Score: 432
Average Score: 432.0
Scores: 432.0
Min Rate: 1/1 (1.00)
Record: Win
macbookair:code warrenroberts
```

Figure 2: Corners Heuristic Results

Why This Heuristic Works This heuristic is a lower bound because:

- Manhattan distance ignores walls, so it never overestimates the true maze path,
- the MST gives the minimum cost needed to connect all remaining corners,
- and the sum of these values still underestimates or matches the true remaining path cost.

Therefore, the heuristic is admissible. It is also consistent in practice for this problem because each move changes the true cost by at most 1 and the heuristic decreases in a controlled way.

MST Caching To improve efficiency, I cached MST costs inside the problem instance using `problem._mst_cache`. Since the remaining corners are represented as an immutable `frozenset`, they can be used as dictionary keys. This prevents recomputing the same MST many times during * search.

FoodSearchProblem Heuristic

The next task was implementing `food_heuristic(state, problem)` for `FoodSearchProblem`. In this problem, Pacman must collect every food dot in the maze.

State Representation The provided state representation is:

Pacman position food grid)

where the food grid stores which food dots remain.

Heuristic Idea My heuristic computes:

1. the Manhattan distance from Pacman to the nearest food,
2. plus the maximum Manhattan distance between any two remaining food dots.

Formally:

$$h(n) = \text{distance to nearest food} + \text{food diameter}$$

The nearest-food term estimates the cost to reach the food region, while the diameter term estimates the spread of the remaining food once Pacman gets there.

Why This Heuristic Is Reasonable This heuristic is useful because:

- Pacman must reach at least one remaining food dot,
- and if the food dots are spread apart, Pacman must eventually cover that spread.

Since Manhattan distance ignores walls, both parts are optimistic estimates, keeping both terms as optimistic distance estimates. In testing, this heuristic significantly improved * performance compared to uninformed search.

```
macbookair:code warrenroberts python3 pacman.py -l tricky_search -p SearchAgent -a fn=astar,prob=FoodSearchProblem,heuristic=
food_heuristic
[SearchAgent] using function astar and heuristic food_heuristic
[SearchAgent] using problem type FoodSearchProblem
Path found with total cost of 63 in 0.8 seconds
Search nodes expanded: 7788
Your Pacman Hits Walls 1 Times
Pacman emerges victorious! Score: 568
Average Score: 568.0
Scores: 568.0
Min Rate: 1/1 (1.00)
Record: Win
macbookair:code warrenroberts
```

Figure 3: Eating all food results

ClosestDotSearch gent

The final task was implementing the greedy search strategy in:

```
find_path_to_closest_dot(self, game_state)
```

approach I used `nyFoodSearchProblem`, which treats any location containing food as a goal state. Then I used BFS to find a shortest path from Pacman's current position to the nearest food dot.

This process is repeated until no food remains:

- find the nearest food,
- move Pacman there,
- update the game state,
- repeat.

Wall-Hit Handling Because the closest-dot problem involves many repeated searches during one full run, I ensured that the intentional wall hit occurs only once overall by tracking:

```
self.num_hits
```

inside `ClosestDotSearch gent`. On the first segment only, the agent attempts a controlled wall bonk using an adjacent wall and then proceeds with BFS to the closest food.

Why This Is Suboptimal This approach is greedy rather than globally optimal. Choosing the nearest food at each step may lead to a worse overall route than planning for all food at once. However, it is much faster and is appropriate for the assignment's suboptimal-search section.

As a result, intentional wall hits are now part of the expected cost model rather than being treated as outright invalid behavior.

```
macbookair:code warrenroberts$ python3 pacman.py -l big_search -p ClosestDotSearchAgent -z .5
[SearchAgent] using function depth_first_search
[SearchAgent] using problem type PositionSearchProblem
Path found with cost 348.
Your Pacman Hits Walls 2 Times
Pacman emerges victorious! Score: 2362
Average Score: 2362.0
Scores: 2362.0
Win Rate: 1/1 (1.00)
Record: Win
macbookair:code warrenroberts$
```

Figure 4: Suboptimal results

Results

- `CornersProblem` was successfully implemented using a state representation that tracks both position and visited corners.
- `corners_heuristic` used nearest-corner distance plus MST cost to guide * efficiently.
- `food_heuristic` estimated the remaining work using the nearest-food distance and the spread of remaining food dots.
- `ClosestDotSearchAgent` used BFS repeatedly to greedily collect the nearest food until all food was eaten.
- The wall-hit constraint was satisfied in a controlled way without flooding the search space with wall states.

In general, * with a good heuristic performed substantially better than uninformed methods on the harder search tasks, especially when the state space grew to track corners or food configurations.

Conclusion

Assignment 3 extended the search techniques from assignment 2 into more difficult tasks. The key challenge was no longer just finding a path, but designing the right state representation and heuristic.

For the corners problem, tracking visited corners was essential. For * search, the MST-based corners heuristic provided a much better estimate than a trivial heuristic. For the food-search problem, the nearest-food plus food-diameter heuristic gave a practical lower bound that improved search quality. Finally, the greedy closest-dot strategy provided a faster suboptimal alternative when exact search became expensive.

Overall, this assignment demonstrated that the success of search algorithms depends not only on the search method itself, but also on how the problem is structured and the helpfulness of its heuristic.